THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Four cells of 1.5 V each are joined in series. The total voltage is: 1

(a) 1.5 V **(b)** 6.0 V **(c)** 3.0 V **(d)** 4.5 V

2. Which of the following is not a safety measure? 1

(a) using damaged wires **(b)** using rubber mats
(c) using MCB **(d)** using insulated tools

3. A bulb uses a battery of two cells. If a third cell is added in series, the bulb: 1

(a) becomes cold **(b)** goes off **(c)** becomes dimmer **(d)** becomes brighter

4. Two cells of 1.5 V are connected with opposite polarity in series. The net voltage is about: 1

(a) 0 V **(b)** 4.5 V **(c)** 3 V **(d)** 1.5 V

5. An electrician wears rubber gloves because they: 1

(a) are costly **(b)** are heavy **(c)** are conductors **(d)** are insulators

6. A student wants to test an object for conduction. Which item completes the circuit tester? 1

(a) only a switch **(b)** only a bulb
(c) a cell, bulb and wires **(d)** only wires

7. A bulb does not glow in a circuit although the cell is new. The most likely reason is: 1

(a) the filament is broken or the circuit is open
(b) the switch is closed
(c) the cell is full
(d) the wires are copper

8. Which arrangement correctly connects a cell and bulb? 1

(a) each bulb terminal connected to a different cell terminal
(b) both bulb terminals to the same cell terminal
(c) no wires
(d) only one terminal connected

9. Pencil lead (graphite) in a circuit tester makes the bulb: 1

(a) break **(b)** glow **(c)** stay off **(d)** melt

10. A student connects a cell, switch and bulb in a loop. The bulb glows only when: 1

(a) the switch is closed **(b)** the bulb is covered
(c) the switch is open **(d)** the cell is removed

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A student builds a tester with a cell, bulb and two wires. The bulb glows when a coin is placed but not with a plastic cap. Explain. 2

12. Why are cells not connected to opposite polarity in series? 2

13. What precaution should you take while replacing a fuse? 2

14. Why is a coiled heating element better than a straight wire in a heater? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Four 1.5 V cells are joined in series to light a 6 V bulb. Does the bulb glow normally? Explain. 3

16. Describe how an MCB works in simple terms. 3

17. Describe a safe way to disconnect a plugged-in device. 3

OR

Describe how you would design a simple circuit tester and test five objects. 3

18. Explain why we use thick wires for geysers and thin wires for bulbs. 3

OR

Why is graphite used in the lead of pencils and also in electrodes? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe a complete experiment to find which materials are conductors, including the circuit diagram in words, table of results and conclusion. 5

OR

Describe how an electrician stays safe while repairing a socket. 5

20. Explain the heating effect of current with a simple experiment using a nichrome wire and cells. 5

OR

Explain what happens in a short circuit and why it is dangerous. 5

END OF QUESTION PAPER — Total: 40 Marks

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